

MSE 312: Mechatronics Design II

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Dr Patrick Palmer, Professor SFU



- Electrical Engineer P.Eng, C.Eng
- Founded 3 companies
- Consulted for many top companies
- Previously Faculty at the University of Cambridge, UK and UBC
- Emeritus Fellow of St Catharine's College Cambridge
- Taught MSE 312 in 2022, 2023, 2025



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Recap

Kinetic Mechatronic Mechanisms



Designing the mechanical foundation for a controllable one-degree-of-freedom system

Design and build a motor-driven mechanical launcher that moves a small ball/projectile from a starting position to a target using a **truss-embedded mechanical mechanism**.

The design must include analytical mechanism modeling, CAD, physical prototyping, and experimental validation.

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Recap

MSE312 Project

Mechanical project goal:

Design a system that uses one primary rotary input to produce a controlled, repeatable visible output motion.

Project examples:

Crank-slider, four-bar linkage, or cam/follower.

Driving question:

How can a mechanical design transform motor input into useful, repeatable, and controllable motion while minimizing inertia, deflection, failure risk, and fabrication uncertainty?

What can you control?

motion path, mass distribution, stiffness, friction/backlash, joint quality, sensor mounting, speed range



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Recap

WEEK 1



Project constraints

- One primary actuated input shaft or motor axis.
- The design must use one or two mechanisms selected from **slider-crank, cam mechanism, and four-bar linkage**.
- Gears are optional, with a **maximum of one gear pair**.
- Passive links, joints, sliders, followers, guides, and supports are allowed.
- At least one moving link must include a truss-embedded or lightweight skeletal structure.
- Use course-approved materials: brass rod, plates, fasteners, solder, and limited 3D-printed plastic.
- Fit within the course envelope and fixture requirements.
- Teams must perform the course-defined launching task and define a clear measurement method.



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Recap

WEEK 1



What makes this a mechatronic design project?

- The motor does not simply “make it move”; the mechanical architecture shapes the control problem.
- A nonlinear linkage may require different control logic than a simple arm.
- A high-inertia mechanism may need more torque, slower acceleration, and gentler control.
- A flexible mechanism may create overshoot, vibration, and poor repeatability.
- A mechanism with backlash, friction, or jamming may fail even with good electronics.



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What this Course IS

Definition of **engineer** noun from the Oxford Advanced Learner's Dictionary

engineer *noun*



/,endʒɪ'nɪə(r)/

/,endʒɪ'nɪr/

- 1 ★ a person whose job involves designing and building engines, machines, roads, bridges, etc.

- We worked with a team of scientists and engineers.
- You need the advice of a qualified engineer.
- engineers who designed and built advanced military aircraft

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What this Course IS NOT



Archimedes
doing
calculations as he
is killed

Not a Hackathon!

<https://interestingengineering.com/13-engineers-from-antiquity-and-their-marvels>

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Integration Objectives

- Integrate mechanical, electrical, sensors and controls knowledge to solve a design problem.
- Design and operate a basic dc motor drive, interface electronics, and electrical system for a mechatronics application
- Understand the role and use of various sensors, build a feedback control system and apply it to a mechatronic system
- Consider safety

Project Based Learning

“Students learn about a subject by working for an extended period of time to investigate and respond to a complex question, challenge or problem.”

Buck Institute for Education (PBLWorks)

It needs to be authentic and engaging

Assessment

Evaluation	Percentage of overall grade
Mechanical part (labs + presentation)	33%
Electronics/control part	33%
Integration part	34%

Integration Part

- A Demonstration/ Competition
- Documentation
- A final INDIVIDUAL Reflection

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Health and Safety

- You must observe all required health and safety information. This includes both Covid-19 and Lab safety.
- Obtain and wear safety glasses
- Wash hands often, use hand sanitiser!
- Masks may be required.

If you are feeling poorly, stay home and interact with your team using Whatsapp, Zoom etc.

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The Ball and Playing Area

- Balls
 - The projectiles used will be a cedar ball, a bouncy ball and a fluffy ball. The ball weights are approximately 2 grams and each has a diameter of approximately 20 mm. They are colored appropriately.
- Playing Area
 - A pair of vertical mounting rails will be provided on a base.
 - A DC motor driven shaft will be attached to the rails provided.
 - You may position an arm of your own design at a height of your choosing.

BOOKABLE playing areas will be made available for final tests.



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Competition Date and Etiquette 2026

Date: Friday 7th August 2.30 – 7.30

- **PLEASE PUT THIS IN YOU CALENDAR NOW**
 - 2.30 – 4.30 (class time) priority is for Wednesday and Thursday Groups
 - A Booking system will be used
- You are expected to be around for the whole session, if at all possible
 - It will help you write your final Individual Reflection Report



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The Balls and Playing Area

- Balls: Cedar wood balls and Translucent bouncy balls
 - The cedar balls weigh 2g* with a diameter of 20mm*
 - The translucent bouncy balls weigh 6.5g* with a diameter of 24mm*
- Ball receiving
 - Balls will appear one at a time to a fixed position (set by your team)
 - Balls will appear in random order
- Playing Area
 - A perforated metal mounting plate for your mechanism on your bench.
 - A moveable 1/12 scale* basketball hoop (internal rim diameter 42mm*)
 - The backboard of the hoop shall be between 1 and 2 m from the furthest point the centre of a ball can be held level in your mechanism when in throwing motion
(to be measured in the competition)



* dimensions are approximate and measurement is recommended

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The Mounting Plate and Hoop



- The Mounting Plate sits on the bench
 - Your mechanism can be mounted at any height above the perforated plate*
 - Fixing to the plate is required!
- The Hoop sits on the bench, with its base approximately level with the mounting plate

* In consultation with the instructor team and depending on the safety and approval of the method, the mounting plate can be raised above the bench.

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Performance Objectives

- **Attribute 1: Accuracy:** Can you get a ball into the hoop that is placed at a given distance? You will be informed of the distance later in the Semester (well in advance of your demonstration).
- **Attribute 2: Robustness:** Can your system accommodate two different balls (size, weight and type) while maintaining accuracy, possibly with differing trajectories.
- **Attribute 3: Autonomy:** Can your system detect the type of ball and determine the required inputs for an accurate throw and repeat this process without intervention.
- **The scoring system* will capture the above appropriately:** Scoring a 'bank shot' will gain a bonus, and scoring a 'swish shot' will also gain a bonus. Hitting the backboard and failing to enter the hoop will score according to where the ball strikes the backboard markings. Your total demonstration score is a combination of successfully completing the trials in competition with other teams (pre-trial presentation, ball score, time, style, robustness, etc.).

*Further details regarding the competition rules will be provided nearer the time



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Integration of Mechatronic Systems

- Review of the Design Process of the whole system
 - Choice of Mechanism and trade-offs
 - Trajectory planning,
 - Ball detection method
 - Sensor choice, interfaces and system operation
- Data - Measuring and recording at the Competition
 - Score based on various factors
- Evaluation of whole system
 - Quality of Manufacture of the integrated system/controls
- Safety and Risk Mitigation



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INTEGRATION MARK BREAKDOWN

Component	Activity	Marks	Comment
Presentation prior to Demonstration	Team	5	Team members talk and answer questions (Videod)
Quality	Create a working system	5	Not a Hackathon
Documentation	Team	4	Final Design Documents for review
Demonstration/Competition		15	Each shot videoed. To be in the competition requires rule following
Individual Reflection	Individual!	5	On the whole process and Teamworking

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Demonstration vs Competition

- Combined Demonstration and Competition
- Score includes
 - Ball detection
 - Score for each shot
 - Forward motion
 - In the Hoop
 - Hit the backboard
 - Style Bonus

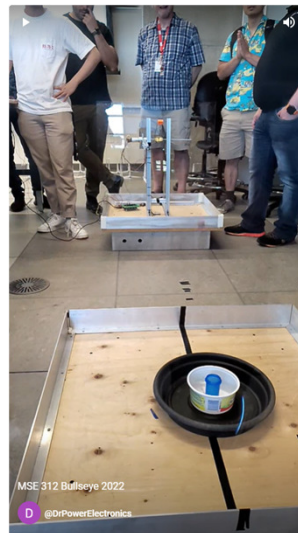
NOT A HACKATHON!

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Any Questions?

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2022



https://www.youtube.com/shorts/QM5_liLYeEU

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Teamworking skills

- Communication
- Conflict Management
- Task Management
- Effective Meetings

A team has a range of skills, multiple independent thinkers, multiple amounts of time and multiple viewpoints.

Don't waste these resources!



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Contributions

- Social Loafers (“hitchhikers”, freeloaders) count on your self-sacrifice to take advantage of you
- Contribution is not measured by ‘noise’ or most ‘productive’
- EGBC Code of Ethics (*New 2021*)
 - 11. clearly identify each registrant who has contributed professional work, including recommendations, reports, statements, or opinions;



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Project skills you need for success

- Technical skills
 - Proficiency in core and supporting technology areas
- Project management skills
 - A plan, timelines, communication plan
- Teamworking Skills
 - Interpersonal & behavioral skills
 - Relationship building, negotiating, influencing, delegating and conflict resolution



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Communication tips for Great Communication

- Practice communication skills
 - Listen well
 - Observe carefully
 - Give feedback constructively
- Consider the function of the communication
- We cannot **not** communicate
- Use the most appropriate method
 - Phone (voip), an email (chat message), a meeting
- Use your writing skills
- Follow up
- Consider the long-term implications

"This could've been an email"



<https://ahseeit.com/?qa=63963/this-couldve-been-an-email-meme>



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